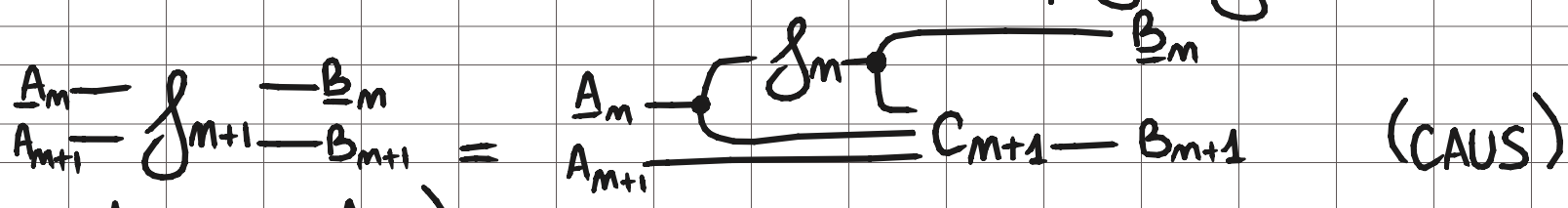
A NOTE ON STATIONARY PROCESSES

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- ongoing work with Yoàv Montacute

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Consider a copy-discard category $\mathcal{C}$ with conditionals and the copy-discard category $\mathbf{Proc}_\mathcal{C}$ of causal processes over it. Its objects are streams of objects of $\mathcal{C}$, $\mathbb{A} = (A_0, A_1, A_2, \dots)$, and its morphisms $f : \mathbb{A} \to \mathbb{B}$ are streams of morphisms $f_n : A_0 \otimes \cdots \otimes A_n \longrightarrow B_0 \otimes \cdots \otimes B_n$ in $\mathcal{C}$ such that, for all $n \in \mathbb{N}$, there is a morphism $c_{n+1} : \underline{B}_n \otimes \underline{A}_n \otimes A_{n+1} \longrightarrow B_{n+1}$ satisfying



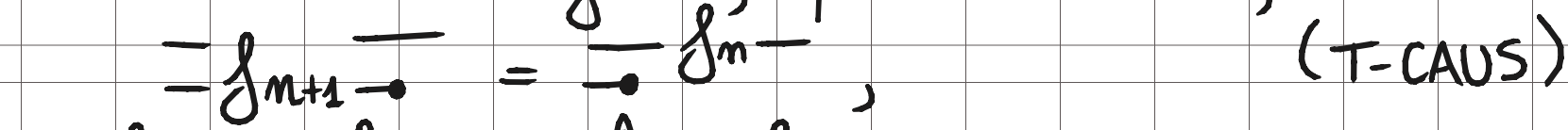
(CAUS)

($\underline{A}_n := A_0 \otimes \cdots \otimes A_n$).

The example we keep in mind is $\mathcal{C} = \mathrm{Kl}\,\mathcal{D}$ the Kleisli category of the finitely (or countably) supported distribution monad on Set. We may consider Kleisli categories of other monads, like the subdistributions, powerset, nonempty powerset or Lévy monads.

We define stationary and exponentially α-mixing causal processes, hopefully generalizing the existing definitions for stochastic processes.

For now, we will assume that all morphisms are total, $-f- \bullet = - \bullet$; this simplifies the condition (CAUS) to just, for all $n \in \mathbb{N}$,



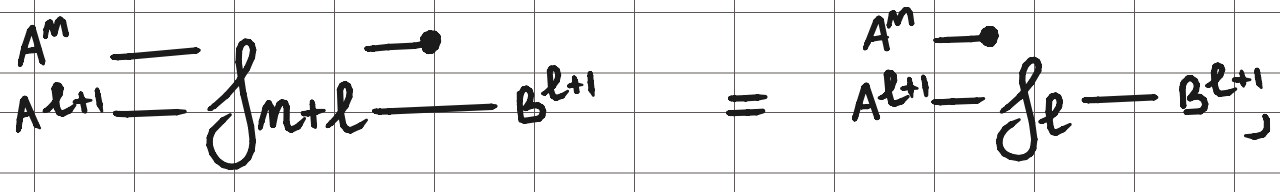
(T-CAUS)

i.e. $f_{n+1} ; \pi^B_{0:n} = \pi^A_{0:n} ; f_n$.

We can probably remove this restriction by carrying around the "probabilities of failure".

STATIONARY PROCESS

A process $f : A^\omega \to B^\omega$ is stationary if, for all $n, \ell \in \mathbb{N}$,



$$(STA)$$

i.e. $f_{n+\ell} ; \pi^B_{n:n+\ell} = \pi^A_{n:n+\ell} ; f_\ell$.

ex For $\mathcal{C} = \mathrm{Kl}\,\mathcal{D}$ and $A = 1$, we should obtain the definition of stationarity for stochastic processes.

LEMMA 1

Stationary processes form a monoidal subcategory of $\mathrm{Proc}_\mathcal{C}$. It also contains all the "memoryless" processes, so it is a copy-discard subcategory.

PROOF

Consider two stationary processes, $f : A^\omega \to B^\omega$ and $g : B^\omega \to C^\omega$; their composition is also stationary:

$$\begin{aligned}
&(f;g)_{n+\ell} ; \pi^C_{n:n+\ell} \\
&:= f_{n+\ell} ; g_{n+\ell} ; \pi^C_{n:n+\ell} \\
&= f_{n+\ell} ; \pi^B_{n:n+\ell} ; g_\ell && (g \text{ stationary}) \\
&= \pi^A_{n:n+\ell} ; f_\ell ; g_\ell && (f \text{ stationary}) \\
&=: \pi^A_{n:n+\ell} ; (f;g)_\ell
\end{aligned}$$

Similarly, for two stationary processes $f : A^\omega \to B^\omega$ and $f' : A'^\omega \to B'^\omega$, their monoidal product is also stationary:

$$\begin{aligned}
&(f \otimes f')_{n+\ell} ; \pi^{B\otimes B'}_{n:n+\ell} \\
&:= \sigma^{n+\ell}_{A,A'} ; (f_{n+\ell} \otimes f'_{n+\ell}) ; \sigma^{-(n+\ell)}_{B,B'} ; \pi^{B\otimes B'}_{n:n+\ell} \\
&\overset{(NAT)}{=} \sigma^{n+\ell}_{A,A'} ; (f_{n+\ell} \otimes f'_{n+\ell}) ; (\pi^B_{n:n+\ell} \otimes \pi^{B'}_{n:n+\ell}) ; \sigma^{-\ell}_{B,B'} \\
&\overset{(MON)}{=} \sigma^{n+\ell}_{A,A'} ; ((f_{n+\ell} ; \pi^B_{n:n+\ell}) \otimes (f'_{n+\ell} ; \pi^{B'}_{n:n+\ell})) ; \sigma^{-\ell}_{B,B'} \\
&\overset{(STA)}{=} \sigma^{n+\ell}_{A,A'} ; ((\pi^A_{n:n+\ell} ; f_\ell) \otimes (\pi^{A'}_{:n+\ell} ; f'_\ell)) ; \sigma^{-\ell}_{B,B'} \\
&\overset{(MON)}{=} \sigma^{n+\ell}_{A,A'} ; (\pi^A_{n:n+\ell} \otimes \pi^{A'}_{n:n+\ell}) ; (f_\ell \otimes f'_\ell) ; \sigma^{-\ell}_{B,B'} \\
&\overset{(NAT)}{=} \pi^{A\otimes A'}_{n:n+\ell} ; \sigma^{\ell}_{A,A'} ; (f_\ell \otimes f'_\ell) ; \sigma^{-\ell}_{B,B'} \\
&=: \pi^{A\otimes A'}_{n:n+\ell} ; (f \otimes f')_\ell
\end{aligned}$$

where $\sigma^m_{A,B} : (A \otimes B)^{m+1} \longrightarrow A^{m+1} \otimes B^{m+1}$ is the obvious reshuffling of A^{m+1} and B^{m+1}.

Finally, consider a (total) memoryless process, i.e. $f : A^\omega \to B^\omega$ s.t. $f_n = g^{n+1}$ for some (total) $g : A \to B$. Then, f is stationary:

$$\begin{aligned}
&f_{n+\ell} ; \pi^B_{n:n+\ell} \\
&:= g^{n+\ell+1} ; \pi^B_{n:n+\ell} \\
&= (g^m \otimes g^{\ell+1}) ; \pi^B_{n:n+\ell} \\
&\overset{(TOT)}{=} \pi^A_{n:n+\ell} ; g^{\ell+1} \\
&=: \pi^A_{n:n+\ell} ; f_\ell
\end{aligned}$$

Symmetries, associators, unitors, copy and discard are all (total) memoryless processes. □

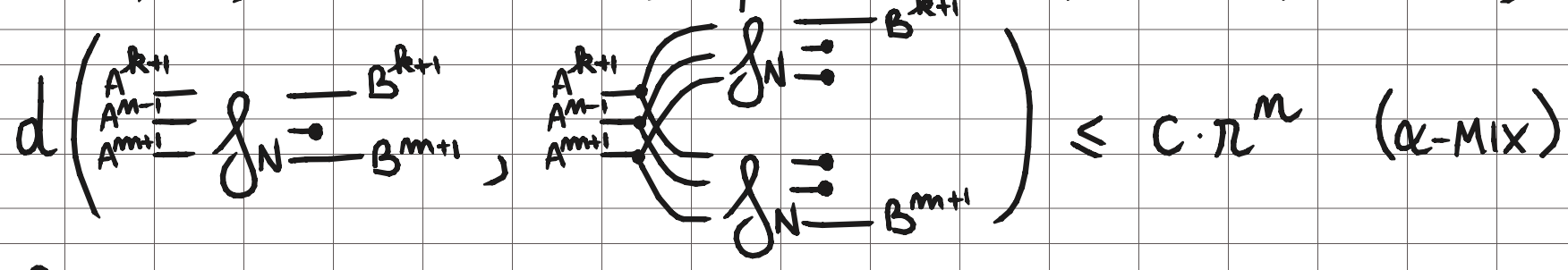
note: I am not sure if stationarity is preserved by feedback as well.

Suppose that the base category $\mathcal{C}$ is, additionally, monoidally $\mathbf{PMet}_T$-enriched.

ex For $\mathcal{C} = \mathbf{KlD}$ we have a $\mathbf{PMet}_1$-enrichment given by the Wasserstein distance, which should be a monoidal enrichment as well. For two morphisms, $f, g : A \to B$, their distance is defined as a fixed point (see [BBKK(2018)] for the definition).

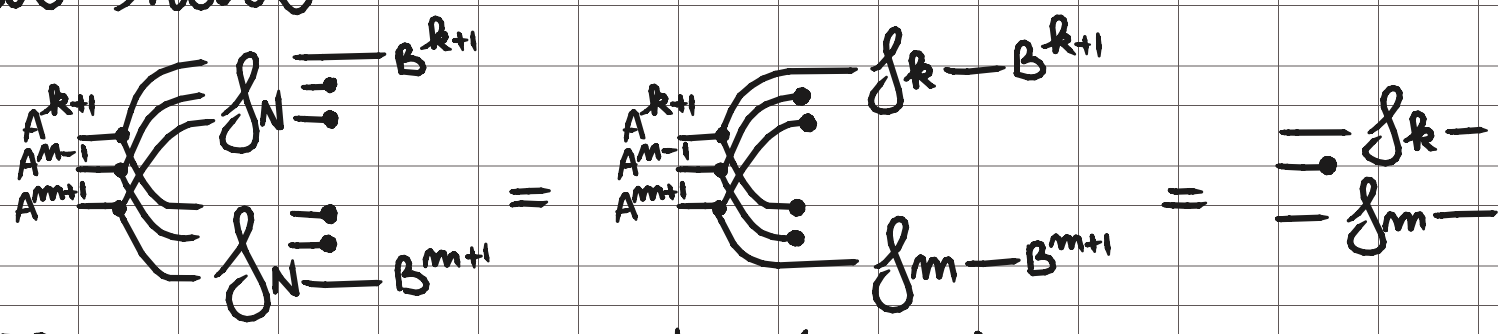
EXPONENTIALLY α-MIXING PROCESS

A stationary process, $f : A^\omega \to B^\omega$, is exponentially α-mixing if there are constants, $c \in \mathbb{R}^+$ and $r \in (0,1)$, such that, for all $k, m \in \mathbb{N}$, $n \in \mathbb{N}^+$,

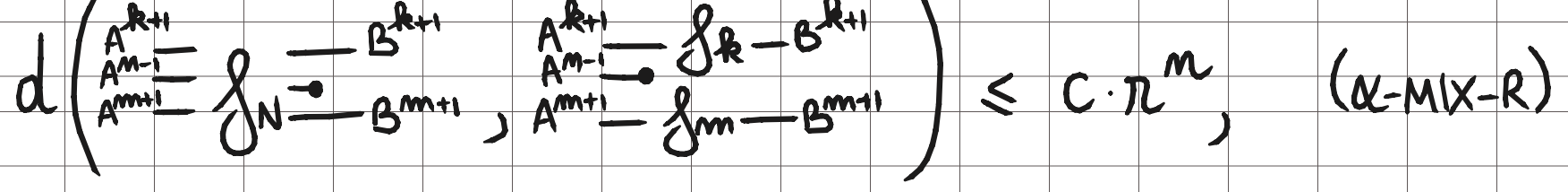


where $N := k+n+m$.

Since f is total, causal and stationary, we have



Then, we can rewrite (α-MIX) as



i.e. $d(f_N ; \pi^B_{0:k,n+k:N}, \pi^A_{0:k,n+k:N} ; (f_k \otimes f_m)) \leq c \cdot r^n$.

note: It is not clear to me that this gives the usual definition of exponentially α-mixing. This definition does produce compositional bounds, but I don't know whether they are practical.

LEMMA 2

Suppose that $f : A^\omega \to B^\omega$, $g : B^\omega \to C^\omega$ and $f' : A'^\omega \to B'^\omega$ are stationary and exponentially α-mixing with constants (c, r), (d, s) and (c', r'), respectively.

Then, $f;g$ and $f \otimes f'$ are exponentially α-mixing with constants $(c+d, \max(r,s))$ and $(c_\otimes, r_\otimes)$, respectively.

PROOF

$$
\begin{aligned}
& d((f;g)_N ; \pi^C, \pi^A ; ((f;g)_k \otimes (f;g)_m) \\
&:= d(f_N ; g_N ; \pi^C, \pi^A ; ((f_k ; g_k) \otimes (f_m ; g_m))) \\
&\overset{(\Delta)}{\leq} d(f_N ; g_N ; \pi^C, f_N ; \pi^B ; (g_k \otimes g_m)) \\
&\quad + d(f_N ; \pi^B ; (g_k \otimes g_m), \pi^A ; (f_k \otimes f_m) ; (g_k \otimes g_m)) \\
&\overset{(ENR)}{\leq} d(g_N ; \pi^C, \pi^B ; (g_k \otimes g_m)) + d(f_N ; \pi^B, \pi^A ; (f_k \otimes f_m)) \\
&\overset{(\alpha\text{-MIX})}{\leq} d \cdot s^n + c \cdot r^n \\
&\leq d \cdot \max(r,s)^n + c \cdot \max(r,s)^n \\
&= (d+c) \cdot \max(r,s)^n
\end{aligned}
$$

$$
\begin{aligned}
& d((f \otimes f')_N ; \pi^{B \otimes B'}, \pi^{A \otimes A'} ; ((f \otimes f')_k \otimes (f \otimes f')_m) \\
&:= d(\sigma^N_{A,A'} ; (f_N \otimes f'_N) ; \sigma^{-N}_{B,B'} ; \pi^{B \otimes B'}, \\
&\qquad \pi^{A \otimes A'} ; (\sigma^k_{AA'} \otimes \sigma^m_{AA'}) ; (f_k \otimes f'_k \otimes f_m \otimes f'_m) ; (\sigma^{-k}_{B,B'} \otimes \sigma^{-m}_{B,B'}))
\end{aligned}
$$

$$
= d\left(
\begin{array}{l}
A^{k+1} \\ A'^{k+1} \\ A^{m-1} \\ A'^{m-1} \\ A^{m+1} \\ A'^{m+1}
\end{array}
\begin{array}{c}
\\ \\ \\ \\ f_N \\ f'_N
\end{array}
\begin{array}{l}
B^{k+1} \\ B'^{k+1} \\ \\ \\ B^{m+1} \\ B'^{m+1}
\end{array}
\quad , \quad
\begin{array}{l}
A^{k+1} \\ A'^{k+1} \\ A^{m-1} \\ A'^{m-1} \\ A^{m+1} \\ A'^{m+1}
\end{array}
\begin{array}{c}
f_k \\ f'_k \\ \\ \\ f_m \\ f'_m
\end{array}
\begin{array}{l}
B^{k+1} \\ B'^{k+1} \\ \\ \\ B^{m+1} \\ B'^{m+1}
\end{array}
\right)
$$

$$
= d\left(
\begin{array}{l}
A^{k+1} \\ A'^{k+1} \\ A^{m-1} \\ A'^{m-1} \\ A^{m+1} \\ A'^{m+1}
\end{array}
\begin{array}{c}
\\ \\ \\ \\ f_N \\ f'_N
\end{array}
\begin{array}{l}
B^{k+1} \\ B'^{k+1} \\ \\ \\ B^{m+1} \\ B'^{m+1}
\end{array}
\quad , \quad
\begin{array}{l}
A^{k+1} \\ A'^{k+1} \\ A^{m-1} \\ A'^{m-1} \\ A^{m+1} \\ A'^{m+1}
\end{array}
\begin{array}{c}
f_k \\ f_m \\ f'_k \\ f'_m
\end{array}
\begin{array}{l}
B^{k+1} \\ B'^{k+1} \\ B^{m+1} \\ B'^{m+1}
\end{array}
\right)
$$

$$
\leq d\left(
\begin{array}{c}
f_N \\ f'_N
\end{array}
\quad , \quad
\begin{array}{c}
f_k \\ f_m \\ f'_k \\ f'_m
\end{array}
\right)
$$

$$
\begin{aligned}
&\leq d(f_N ; \pi^B, \pi^A ; (f_k \otimes f_m)) \\
&\quad * d(f'_N ; \pi^{B'}, \pi^{A'} ; (f'_k \otimes f'_m))
\end{aligned}
$$

$$
\leq (c \cdot r^n) * (c' \cdot r'^n)
$$

$\rightsquigarrow$ I'm not sure if $(*)$ should be $(+)$ or max, but, in either case, we can give a bound of the form $c_\otimes \cdot r_\otimes^n$

□

REFERENCES

[BBKK(2018)] Coalgebraic behavioural metrics (2018) Baldan, Bonchi, Kerstan, König